

AUV Thrust Allocation Procedure

Direction-Scaled Control Allocation

1. Purpose

The objective of the thrust allocation procedure is to convert a desired body-frame wrench into individual thrust commands for the AUV's eight thrusters. The allocator must respect the physical thrust limits of each thruster while producing the best possible commanded wrench.

The current formulation preserves the direction of the wrench while respecting the thrusters' output limits.

2. Notation

The desired body-frame wrench is denoted by $\boldsymbol{\tau}_d \in \mathbb{R}^6$ and contains the desired forces and torques along/about the AUV body-frame axes:

$$\boldsymbol{\tau}_d = [F_x \ F_y \ F_z \ \tau_x \ \tau_y \ \tau_z]^T$$

The thrust command vector is denoted by $\mathbf{u} \in \mathbb{R}^8$, where each element corresponds to the force command sent to one thruster:

$$\mathbf{u} = [u_1 \ u_2 \ u_3 \ u_4 \ u_5 \ u_6 \ u_7 \ u_8]^T$$

The allocation matrix is denoted by $\mathbf{T} \in \mathbb{R}^{6 \times 8}$. It maps individual thruster forces to the net wrench applied to the vehicle:

$$\mathbf{T} \mathbf{u} = \boldsymbol{\tau}$$

Each column of $\mathbf{T}$ represents the contribution of one thruster to the six wrench components. More details on the allocation matrix can be found in the Propulsion package README of the AUV-2026 GitHub repository.

3. Basic Problem Formulation

Ideally, we want to find a thrust vector $\mathbf{u}$ that exactly produces the desired wrench:

$$\mathbf{T} \mathbf{u} = \boldsymbol{\tau}_d$$

At the same time, every thruster must remain within its physical force limits:

$$u_{\min} \leq u \leq u_{\max}$$

For the T200 thrusters used on the AUV, the thrust bounds are asymmetric. For example, a representative bound is:

$$u_{\min} = -30 \text{ N}, \quad u_{\max} = 40 \text{ N}$$

More generally, each thruster may have its own minimum and maximum thrust limits.

If the desired wrench is feasible, then there exists at least one thrust vector $\mathbf{u}$ satisfying both the wrench equality and the thrust bounds. Because the AUV is over-actuated, there may be infinitely many thrust vectors that generate the same wrench. In that case, we prefer the solution with the smallest magnitude to reduce unnecessary actuator effort and power consumption.

Minimize:

$$\frac{1}{2} \|\mathbf{u}\|^2$$

subject to:

$$\mathbf{T} \mathbf{u} = \boldsymbol{\tau}_d$$

$$u_{\min} \leq \mathbf{u} \leq u_{\max}$$

4. Handling Infeasible Wrenches

Some desired wrenches may not be physically achievable within the thruster limits. For example, the controller may request more surge force, yaw torque, or combined force/torque than the thrusters can produce.

A simple but poor approach would be to compute the pseudoinverse solution and clip individual thruster outputs. However, clipping can distort the resulting wrench direction because each thruster is saturated independently after the allocation has already been computed.

Instead, we introduce a scaling factor $\alpha \in [0, 1]$. The value of α represents the largest achievable fraction of the desired wrench while respecting the thrust limits.

$$\mathbf{T} \mathbf{u} = \alpha \boldsymbol{\tau}_d$$

If $\alpha = 1$, the full desired wrench is feasible. If $\alpha < 1$, the allocator produces the largest possible wrench in the same direction as the desired wrench.

5. Step 1: Maximize the Achievable Wrench Fraction

The first step is to find the largest value of α that can be achieved without violating the thruster limits. This is formulated as a linear program.

The design variable is:

$$\mathbf{x} = [\mathbf{u}^T \ \alpha]^T \in R^9$$

The objective is to maximize α . Since most solvers are written in minimization form, this is written as:

Minimize:

$$f(\mathbf{x}) = -\alpha$$

The constraints are:

$$\mathbf{T} \mathbf{u} = \alpha \boldsymbol{\tau}_d$$

$$u_{\min} \leq \mathbf{u} \leq u_{\max}$$

$$0 \leq \alpha \leq 1$$

After solving this linear program, we extract the optimal value α^* . For a physically feasible wrench, α^* should be equal to 1, or extremely close to 1. If $\alpha^* < 1$, the full desired wrench is not achievable, so the allocator produces the largest feasible scaled version.

6. Step 2: Minimize Thruster Effort

Once α^* has been found, we construct the target wrench. First, we scale the full desired wrench:

$$\boldsymbol{\tau}_{\text{target}} = \alpha \boldsymbol{\tau}_d$$

The second step is to find the smallest-magnitude thrust vector that produces this target wrench:

Minimize:

$$\frac{1}{2} \|\mathbf{u}\|^2$$

subject to:

$$\mathbf{T} \mathbf{u} = \boldsymbol{\tau}_{\text{target}}$$

$$\mathbf{u}_{\min} \leq \mathbf{u} \leq \mathbf{u}_{\max}$$

This is a quadratic program because the objective function is quadratic and the constraints are linear. After solving this problem, we obtain the final thrust vector $\mathbf{u}^*$, which is converted into PWM commands and sent to the thrusters.

7. Practical Computational Shortcut

In practice, the second-stage quadratic program does not always need to be solved.

First, we compute a candidate solution using the Moore–Penrose pseudoinverse:

$$\mathbf{u}_{\text{pinv}} = \mathbf{T}^\dagger \boldsymbol{\tau}_{\text{target}}$$

This gives the minimum-norm solution to the unconstrained system:

$$\mathbf{T} \mathbf{u} = \boldsymbol{\tau}_{\text{target}}$$

The pseudoinverse solution is extremely fast to compute. Therefore, we first check whether it satisfies the thrust bounds:

$$\mathbf{u}_{\min} \leq \mathbf{u}_{\text{pinv}} \leq \mathbf{u}_{\max}$$

If the pseudoinverse solution respects the bounds, we accept it directly. If it violates any thrust limit, we solve the second-stage quadratic program to find the minimum-norm solution that also satisfies the actuator constraints.

8. Final Practical Algorithm

1. Solve a linear program to find the maximum feasible scaling factor α^* .
2. Construct the target wrench by scaling the wrench by α^* .
3. Compute a candidate thrust vector using the Moore–Penrose pseudoinverse.
4. If the pseudoinverse solution respects the thrust limits, use it directly.
5. Otherwise, solve the bounded quadratic program.
6. Convert the final thrust vector into PWM commands and publish them to the thrusters.

9. Summary

This allocation procedure preserves the direction of the wrench , respects thruster limits, and avoids unnecessary QP solves whenever the pseudoinverse solution is already feasible. The result is a practical and efficient thrust allocator suitable for the AUV propulsion package.